

Medium-Range Structural Properties of Vitreous Germania Obtained through First-Principles Analysis of Vibrational Spectra

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We analyze the principal vibrational spectra of vitreous GeO_2 and derive therefrom structural properties referring to length scales beyond the basic tetrahedral unit. We generate a model structure that yields a neutron structure factor in accord with experiment. The inelastic-neutron, the infrared, and the Raman spectra, calculated within a density-functional approach, also agree with respective experimental spectra. The accord for the Raman spectrum supports a Ge-O-Ge angle distribution centered at 135° . The Raman feature X_2 is found to result from vibrations in three-membered rings, and therefore constitutes a distinctive characteristic of the medium-range structure.

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Vitreous germania ($\nu\text{-GeO}_2$) not only plays a prominent role in the optical fiber technology [1] but has also been studied extensively as an archetypical material in glass science [2,3]. Vitreous germania shares with vitreous silica ($\nu\text{-SiO}_2$) a structure consisting of a continuous random network of corner-sharing tetrahedra, with the cation at their center and the O atoms at their corners. However, the packing density of tetrahedra, an adimensional property defined by rescaling the atom density by the cation-oxygen distance, is higher in $\nu\text{-GeO}_2$ [4]. This difference is reflected by the average intertetrahedral bond angle, which is found to be significantly lower in $\nu\text{-GeO}_2$ (133° [5]) than in $\nu\text{-SiO}_2$ (151° [5,6]). Furthermore, the higher packing density in $\nu\text{-GeO}_2$ is also expected to lead to a different ring statistics, in which three- and four-membered rings with small intertetrahedral angles are preferred [7]. However, little is known about such rings in $\nu\text{-GeO}_2$.

It is notoriously difficult to access structural properties beyond the basic structural unit in glassy systems [8]. This should be contrasted with the easy availability of vibrational spectra, which could potentially provide a wealth of information on such medium-range structural aspects. However, an approach based on the interpretation of vibrational spectra can be pursued only through accurate modeling, because of the complexity of the involved vibrations and coupling factors. For instance, the Raman spectrum in $\nu\text{-SiO}_2$ is particularly informative on both the intertetrahedral angles and the small ring structures. Indeed, dramatic changes were observed in the simulated Raman spectrum for models with different angular distributions [9]. Furthermore, two sharp lines in the spectrum are signatures of small ring structures embedded in the network [10]. The Raman spectrum of $\nu\text{-GeO}_2$ is expected to carry similar information. However, despite the likely higher concentration of small rings, this spectrum does not show any sharp feature, being dominated by a broad peak with two shoulders of uncertain origin (X_1 and X_2) [11,12]. Also, it is not clear what kind of information this spectrum provides on the Ge-O-Ge angle distribution.

In this work, we investigate the vibrational spectra of vitreous GeO_2 from first principles, focusing, in particular, on the Raman spectrum and on its relation to the atomic structure beyond the basic tetrahedral unit. Using classical and first-principles methodologies in sequence, we generate a model structure and find good agreement with neutron diffraction data [3]. The vibrational properties of our model are then validated through successful comparisons between calculated and measured inelastic-neutron [13,14] and infrared [11] spectra. We also find good agreement with experiment for the Raman spectrum, thereby supporting a Ge-O-Ge angle distribution centered at 135° . We find that the Raman intensity originating from small rings is very broad, indicating strong couplings between ring and network vibrations. Three-membered rings significantly contribute to the shoulder X_2 , while the feature X_1 results from network motions which are not specifically related to small rings.

The electronic structure in this work is treated within a generalized gradient approximation (GGA) [15] to density functional theory (DFT). Core-valence interactions are described through a norm-conserving pseudopotential for Ge [16] and an ultrasoft one for O [17]. For the electron wave functions and charge density, we used plane-wave basis sets defined by energy cutoffs of 24 and 250 Ry, respectively [18]. The Brillouin zone is sampled at the Γ point. We derived the vibrational frequencies and eigenmodes from the dynamical matrix, which was calculated numerically by taking finite differences of the atomic forces [19]. For accessing the infrared and Raman spectra, we take advantage of a recently developed scheme for applying a finite electric field in periodic cell calculations [20]. We obtained the relevant coupling tensors by numerically calculating first and second derivatives of the atomic forces with respect to the electric field [21]. We applied fields of ± 0.01 a.u.

To generate a model structure of $\nu\text{-GeO}_2$, we first carried out classical molecular dynamics [22] of SiO_2 at the experimental packing density of tetrahedra in $\nu\text{-GeO}_2$.

Through a quench from the melt, we generated a disordered structure consisting of a chemically ordered network of corner-sharing tetrahedra. A model of ν -GeO₂ was then obtained by rescaling the simulation cell by the Ge-O/Si-O bond length ratio and by further optimizing the atomic positions through damped first-principles molecular dynamics [18,23]. The final model consists of 168 atoms in a periodically repeated cubic cell at the experimental density of ν -GeO₂ (3.65 g/cm³ [4]).

The average Ge-O bond length (1.78 Å) is found to be only slightly larger than the experimental one (1.74 Å [4]), in accord with a general tendency of the GGA [16,24]. The structure shows well defined tetrahedral units with an average O-Ge-O angle close to the ideal one and a standard deviation of 6°. The Ge-O-Ge angle distribution has an average angle of 135° with a standard deviation of 10.6° (Fig. 4, inset, below), in good agreement with parameters extracted from diffraction data (133°, 8.3°) [5]. Our model contains 4 three- and 5 four-membered rings. The three-membered rings are quasiplanar, as can be inferred from the sum Σ over all bond angles in the ring that averages to 696°, only slightly lower than the ideal value of 720°. Consistently [7], the average Ge-O-Ge angle in these rings (123°) is smaller than the average in the model (135°) (Fig. 4, inset, below). Four-membered rings have an average Ge-O-Ge angle of 136° and do not show any preference for planarity.

Our model structure shows good agreement with experiment [3] for the neutron static structure factor (Fig. 1) [25]. In particular, our model describes well the first sharp diffraction peak at ~ 1.6 Å⁻¹, which originates from the intermediate range order [2]. The slight out-of-phase oscillation observed at high transferred momenta $Q > 7$ Å⁻¹ results from the small difference between theoretical (1.78 Å) and experimental (1.74 Å) bond lengths.

The reliability of our model of ν -GeO₂ also extends to the vibrational properties. In Fig. 2, the calculated neutron vibrational density of states is compared to available experimental data [13,14]. Overall, the comparison is very satisfactory with the theoretical spectrum showing all the

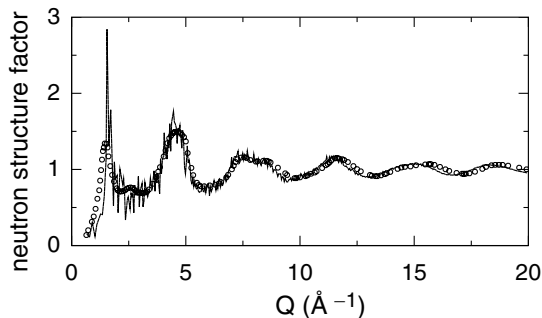


FIG. 1. Calculated (solid line) and measured (circles) [3] neutron static structure factor of ν -GeO₂ at room temperature. We used neutron scattering lengths for Ge and O of 8.193 and 5.805 fm, respectively [33].

salient experimental features. However, the calculated frequencies appear to systematically underestimate the measured ones, an effect that should be attributed to our DFT setup [26]. A detailed analysis [27] shows that the spectrum above ~ 700 cm⁻¹ is mostly given by O stretching motions, with the high-frequency doublet resulting from tetrahedral modes of different symmetry [19]. The vibrational band extending from 380 to 700 cm⁻¹ is composed of O bending motions with a Ge weight close to the atomic fraction, while the lowest band (< 380 cm⁻¹) prevalently features O rocking motions with the Ge/O ratio approaching 1:1 at low frequencies [27].

We calculated a high-frequency dielectric constant of 2.79, only slightly larger than the experimental value (2.58 [28]), as typical in DFT calculations [29]. Anisotropies due to the finite size of the model were found to be $\sim 1\%$. Furthermore, we investigated the imaginary part of the dielectric function ϵ_2 [21,30] and the energy loss function $-\text{Im}(1/\epsilon)$ [21] (Fig. 3). Overall, the comparison between theory and experiment [14] is very good, in particular, for the relative intensities of the three main peaks. The difference between calculated and measured peak frequencies is similar as for the neutron vibrational density of states (Fig. 2). We found longitudinal-optic–transverse-optic splittings of 62, 46, and 150 cm⁻¹ for the three main peaks, in fair agreement with the experimental values of 69, 39, and 116 cm⁻¹ [11,14].

Raman spectra for our model were calculated for incoming and outgoing photons with parallel (HH) and perpendicular polarizations (HV), and compared to experimental spectra [12] in Fig. 4. The theoretical HH spectrum reproduces well the principal peak and the high-frequency doublet, the overall shift to lower frequencies being discussed above. The accord for the HV spectrum is of similar quality. Furthermore, the ratio between the integrated intensities of the HH and HV spectra is closely reproduced in

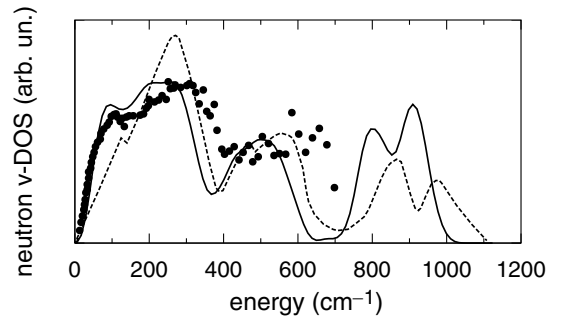


FIG. 2. Calculated (solid line) neutron vibrational density of states (ν -DOS) of ν -GeO₂ at room temperature, compared to corresponding experimental data from Refs. [13] (solid circles) and [14] (dashed line). The experimental data were scaled with respect to the calculated curve, the former to match the height of the highest peak, and the latter to show the same integrated area. In the calculation of the spectrum [19], we used transferred momenta in the range 0.5–4.5 Å⁻¹ [13].

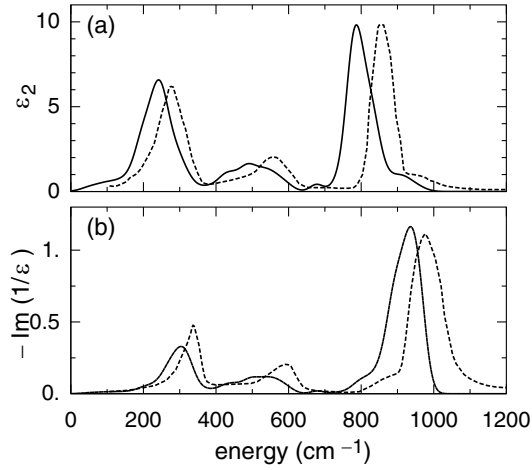


FIG. 3. Comparison between calculated (solid line) and measured [14] (dashed line) infrared spectra of ν -GeO₂, for (a) the imaginary part of the dielectric function and (b) the energy loss function. A Gaussian broadening of 19 cm⁻¹ was used.

our calculation. The calculated shift of the three main peaks upon $^{16}\text{O} \rightarrow ^{18}\text{O}$ substitution [10] also agrees well with experiment [12] (Table I).

The good level of agreement with experiment achieved for the inelastic-neutron and infrared spectra stems from the occurrence of regular tetrahedral units in our model and is only poorly informative on the network organization beyond nearest neighbors [9,21,30]. At variance, the Raman spectrum is highly sensitive to the Ge-O-Ge angle distribution [9]. Indeed, the intensity of the HH Raman spectrum below 700 cm⁻¹ is dominated by O bending

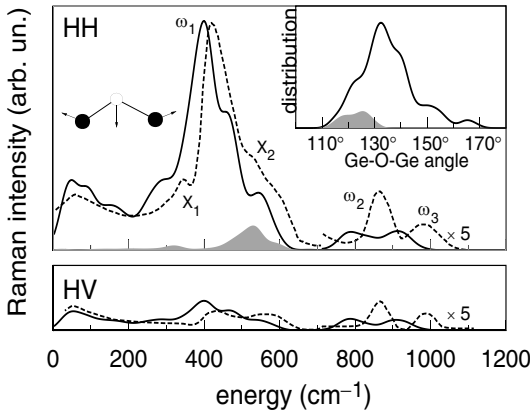


FIG. 4. Calculated (solid line) and measured (dashed line [14]) HH and HV Raman spectra of ν -GeO₂. The calculated spectra are scaled to match the integrated intensity of the experimental HH spectrum. For clarity, the intensities in the high-frequency range were scaled by a factor of 5. Vibrations associated with X_1 are illustrated graphically. Inset: Ge-O-Ge angle distribution in the model structure of ν -GeO₂. Contributions to the HH spectrum and to the angular distribution originating from three-membered rings are shaded. Gaussian broadenings of 19 cm⁻¹ and 2.5° are used.

motions, which account for 93% of the integrated intensity. Hence, the agreement between the shape of the calculated and measured HH Raman spectra provides further support to the Ge-O-Ge angle distribution in our model structure (Fig. 4, inset), in addition to the accord with diffraction data [5] recorded above.

To identify the contribution of three-membered rings to the HH Raman spectrum, we projected the vibrational eigenmodes onto O breathing motions in these rings. These motions correspond to in-phase bending vibrations of the three O atoms in the ring along the bisectors of their respective Ge-O-Ge units [10]. We then determined the contribution of these ring vibrations to the spectrum by using the projected eigenmodes in the expression for the Raman intensities [9]. The Raman projection obtained in this way gives a broad peak centered at 530 cm⁻¹ (Fig. 4), practically coincident with the experimental position of the X_2 shoulder [12]. Upon $^{16}\text{O} \rightarrow ^{18}\text{O}$ substitution, the corresponding projection peak in the theoretical spectra is found to shift by -26 cm⁻¹, in excellent accord with the experimental shift of -25 cm⁻¹ (Table I). These observations support the assignment of the X_2 shoulder to three-membered rings. From the comparison between measured and calculated HH Raman spectra (Fig. 4), we roughly estimate that the concentration of three-membered rings in ν -GeO₂ could reach values twice as high as in our model structure, i.e., involving up to ~20% of the O atoms.

As rationalized by the central-force model [31], the O bending frequency shifts to lower values with increasing Ge-O-Ge angle [31]. Hence, through this relation, the frequency of the X_2 feature can provide information on the degree of planarity of three-membered rings in ν -GeO₂. We would expect that the theoretical frequency for X_2 underestimates the experimental one by about the same amount as the main Raman peak, because these two features result from similar vibrations. Instead, we found close agreement between theory and experiment suggesting that three-membered rings in ν -GeO₂ have larger Ge-O-Ge angles than in our model and correspondingly are more planar. This reasoning is confirmed by the projection onto the most planar ring in our model (with $\Sigma = 707^\circ$), which gives a peak shifted by -23 cm⁻¹ with respect to the experimental X_2 feature, consistent with the shift of the main Raman peak (-18 cm⁻¹).

TABLE I. Calculated isotopic shifts of the main features in the HH Raman spectrum for $^{16}\text{O} \rightarrow ^{18}\text{O}$ substitution, in comparison to respective experimental data [12]. Peak labels are as in Fig. 4. The theoretical shifts of the X_1 and X_2 features were evaluated as described in the text. All shifts are given in cm⁻¹. Experimental errors are given in parenthesis.

	ω_1	ω_2	ω_3	X_1	X_2
Theory	-24	-41	-48	-7	-26
Expt. [12]	-18 (2)	-39 (2)	-45 (2)	-5 (3)	-25 (7)

The width of the projection peak associated with ring vibrations in Fig. 4 is subject to inhomogeneous broadening due to the variety of slightly distorted rings in our model structure. This width is therefore likely to be overestimated because of the residual strain due to the finite size of our model. Thus, we focused only on the Raman projection onto the most planar ring to estimate the width resulting from the coupling to other network vibrations, and found a full width at half maximum (FWHM) of 75 cm^{-1} . This width is significantly larger than the FWHM of 30 cm^{-1} measured for the D_2 line of similar origin appearing in the Raman spectrum of ν -SiO₂ [7,10]. In fact, the typical angle on O atoms in three-membered rings is closer to the average intertetrahedral angle in ν -GeO₂ (133° [5]) than in ν -SiO₂ (151° [5,6]), indicating stronger coupling with network vibrations in the former oxide. This clarifies why the X_2 feature does not give such a sharp peak as the D_2 line of ν -SiO₂.

To elucidate the origin of the feature X_1 appearing at 347 cm^{-1} in the Raman spectrum of ν -GeO₂ [12,32], we first investigated the role of four-membered rings [32] by inspecting the corresponding Raman projections. However, we found that these projections give their largest contribution in correspondence of the principal Raman peak, without any specific relation to X_1 . Then, we analyzed the O bending vibrations throughout the network by considering the associated Ge motions. We found that coupled motions of Ge and O atoms as shown in Fig. 4 give a projection peak (not shown) on the low-frequency side of the main peak. Upon $^{16}\text{O} \rightarrow ^{18}\text{O}$, the feature determined in this way is found to shift by -7 cm^{-1} , in good agreement with the experimental shift of X_1 (-5 cm^{-1} [12]) (Table I). Hence, our analysis supports the assignment of X_1 to diffuse network motions not specifically related to ring structures.

In conclusion, the present investigation of vitreous germania demonstrates the potential of first-principles methods in interpreting vibrational spectra of oxide glasses. More strikingly, such an analysis offers the opportunity to acquire structural information on length scales beyond the basic structural unit, which are generally difficult to access in disordered systems.

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